Fiscal Year:	FY 2020	**Task Last Updated:**	FY 03/10/2020
PI Name:	Mudawar, Issam Ph.D.		
Project Title:	Flow Boiling and Condensation Experiment (FBCE)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	West Lafayette	**State:**	IN
Zip Code:	47907-1288	**Congressional District:**	4
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	06/01/2017	**End Date:**	12/31/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Nahra, Henry K	**Contact Phone:**	216-433-5385
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hasan, Mohammad Ph.D. (NASA Glenn Research Center)		
Grant/Contract No.:	NNX17AK98G		
Performance Goal No.:			
Performance Goal Text:			
	EDITOR'S NOTE: Continuation of "Flow Boiling and Condensation Experiment (FBCE)," grants NNX13AB01G and NNX12AK14G, which covered from 6/1/2012-5/31/2017. Several efforts are presently underway to reduce the size and weight of space systems by transitioning from single-phase to two-phase thermal management. Because of large density differences between liquid and vapor, buoyancy can play an important role in defining the motion of vapor relative to liquid and, therefore, influence heat transfer effectiveness. This is particularly the case for future space missions, including the manned mission to Mars. The emphasis in two-phase fluid physics is evident from several recent NASA workshops that culminated in critical recommendations concerning Rankine cycle power conversion, thermal control systems, and advanced life support systems. Among others, key fluid physics mechanisms that have been identified are flow boiling and condensation. A key strategy is to develop mechanistic models for flow conditions that would ensure insensitivity to gravity level for these important heat transfer		

Task Description:	mechanisms. This would allow performing most of the required confirmatory experiments on Earth. Those topics and the minimum flow requirements constitute key objectives for the proposed study. Unfortunately, most of the two-phase heat transfer know-how amassed over nearly a century of research comes from experiments conducted in Earth's gravity. The proposed study aims to develop an experimentally validated, mechanistic model for the minimum flow criteria to ensure gravity independent flow boiling heat transfer and condensation. Parabolic flight experiments and preparations for International Space Station (ISS) experiments will occur throughout the course of the proposed study. This project will be a joint effort between the Purdue University Boiling and Two-Phase Flow Laboratory (PU-BTPFL) and the NASA Glenn Research Center. Personnel from the two organizations combine extensive experience in research and development of both flow boiling and condensation systems, and in conducting microgravity experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed research will lay the foundation of development of advanced high power to weight ratio thermal management systems for NASA's future space vehicles and planetary bases.
Task Progress:	The present study is focused on development of a 3-D computational approach to predict highly subcooled nucleate flow boiling in vertical upflow. Investigation of existing computational methodology based on Volume of Fluid (VOF) approach revealed fundamental weaknesses in modeling multiphase flows, which stems mostly from inadequate representation of shear-lift force on bubbles. A user defined function is adopted which provides detailed information relating to this important effect, and computed results are validated through comparison with experimental results and analytic predictions of single bubble trajectory. The computational method is then used to model the entire flow field for subcooled flow boiling in a rectangular channel heated on two opposite walls, and predictions are validated against FC-72 experimental data for four different mass velocities. Overall, computationally predicted interfacial behavior, flow pattern, and heat transfer parameters (wall temperature and heat transfer coefficient) show good agreement with experimental data. The model is also shown to be highly effective at predicting local phenomena (velocity and temperature profiles) not easily determined through experiments.
Bibliography Type:	Description: (Last Updated: 01/06/2026)
Abstracts for Journals and Proceedings	Mudawar I, Lee J, O'Neill LE, Hasan MM, Nahra HK, Balasubramaniam R. "Flow Boiling and Condensation Experiment (FBCE) for the International Space Station." 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. ASGSR program and abstracts. 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. , Nov-2019
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Articles in Peer-reviewed Journals	O'Neill LE, Ramaswamy R, Nahra HK, Hasan MM, Mackey JR, Mudawar I. "Identification of condensation flow regime at different orientations using temperature and pressure measurements." International Journal of Heat and Mass Transfer. 2019 Jun;135:569-90. https://doi.org/10.1016/j.ijheatmasstransfer.2019.01.133 , Jun-2019
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Awards	Mudawar I. "Certificate of recognition and celebration of 35 years of service and contributions to Purdue University, December 2019." Dec-2019
Awards	Mudawar I. "American Institute of Aeronautics and Astronautics (AIAA) Space Processing Award, November 2019." Nov-2019
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